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Design and Implementation of a Real-Time Impedance Controller for a 7-DOF Collaborative Robot

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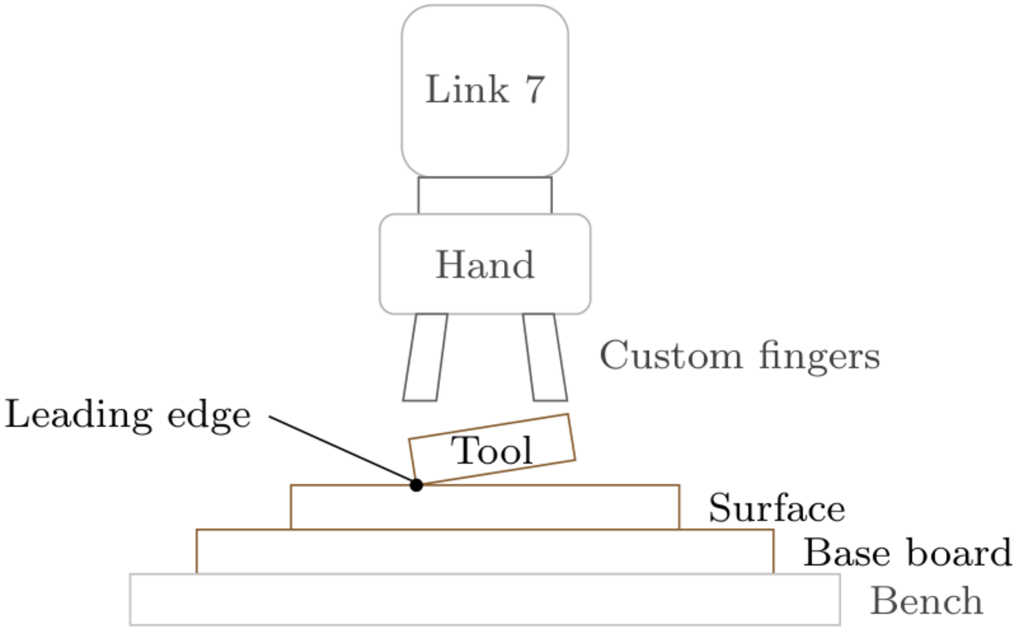
Motivation



Problem

- Tilted tool → edge-first contact
- Rigid tracking resists corrective rotation

Main thesis question

- **How do stiffness and the compliance centre affect passive alignment?**

Idea

- Apply the press; let contact rotate the tool

Overview

1. Cartesian impedance and centre of compliance
2. Contact experiments and results
3. Null-space study
4. Conclusion and future work

Cartesian wrench = force and moment together

Position and force

Desired / measured $p_d, \; p_{\mathrm{EE}}$

$$e_p = p_d - p_{\mathrm{EE}}$$

$$f = K_p e_p + D_p(\dot{p}_d - \dot{p}_{\mathrm{EE}})$$

Orientation and moment

Desired / measured $R_d, \; R_{\mathrm{EE}}$

e_R rotational correction

$$m = K_R e_R + D_R(\omega_d - \omega_{\mathrm{EE}})$$

$K_p, \; K_R$ translational / rotational stiffness

$\dot{p}, \; \omega$ linear / angular velocity

$D_p, \; D_R$ translational / rotational damping

$d, \; \mathrm{EE}$ desired / measured end effector

Real-time control

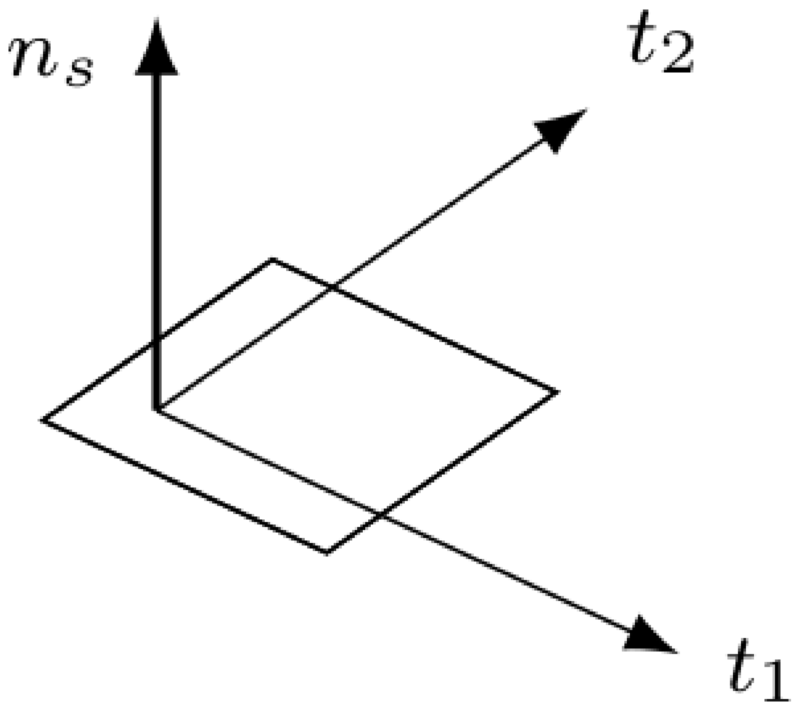
Pose error → Wrench → Joint torques → Robot

Measured pose

$$F = \begin{bmatrix} f \\ m \end{bmatrix}, \qquad \tau_{\text{cart}} = J^{\top}(q)F$$

1 kHz control loop · C++ / libfranka

Surface-relative compliance



- n_s outward surface normal
- $t_1,\ t_2$ two perpendicular surface tangents

Translation

- Softer normal pressing
- Stiffer motion along the surface

Rotation

- Softer rotations about the tangents
- Stiffer rotation about the normal

Centre of compliance (CoC)

Virtual reference point

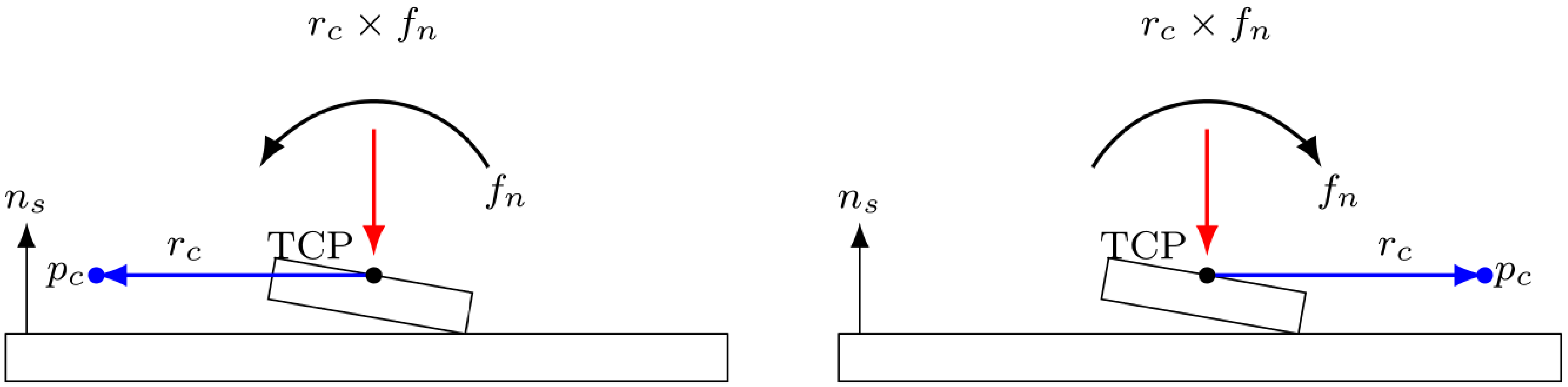
p_c CoC position

r_c Displacement from TCP to CoC

Commanded TCP moment

$$m = m_R + r_c \times f$$

m_R rotational spring-damper moment



(a) Supporting displacement (b) Reversed displacement

Opposite displacement reverses the added moment

Experimental procedure

Tool orientation → **Surface approach** → **Contact establishment** → **Pre-grinding hold**

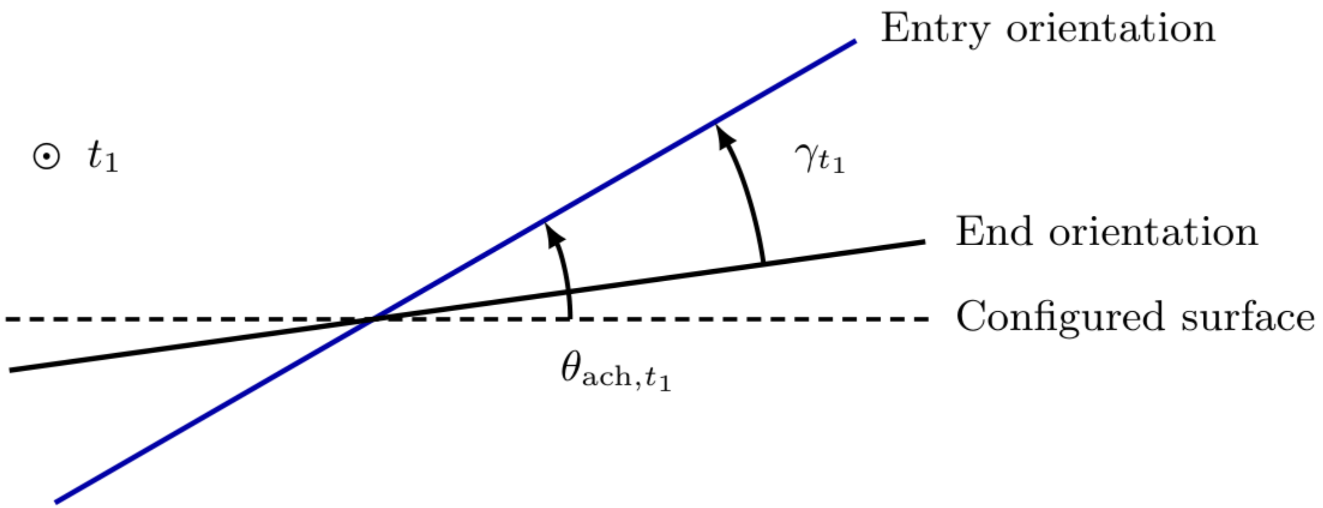
Contact study

- Rectangular tool: 40 × 120 mm
- Five-second contact phase
- Fixed entry orientation reference

Controlled comparison

- A: centre at the TCP
- B / C: rotational / translational stiffness
- D: tangential compliance-centre position
- 57 trials; three repetitions per setting

Angular quantities



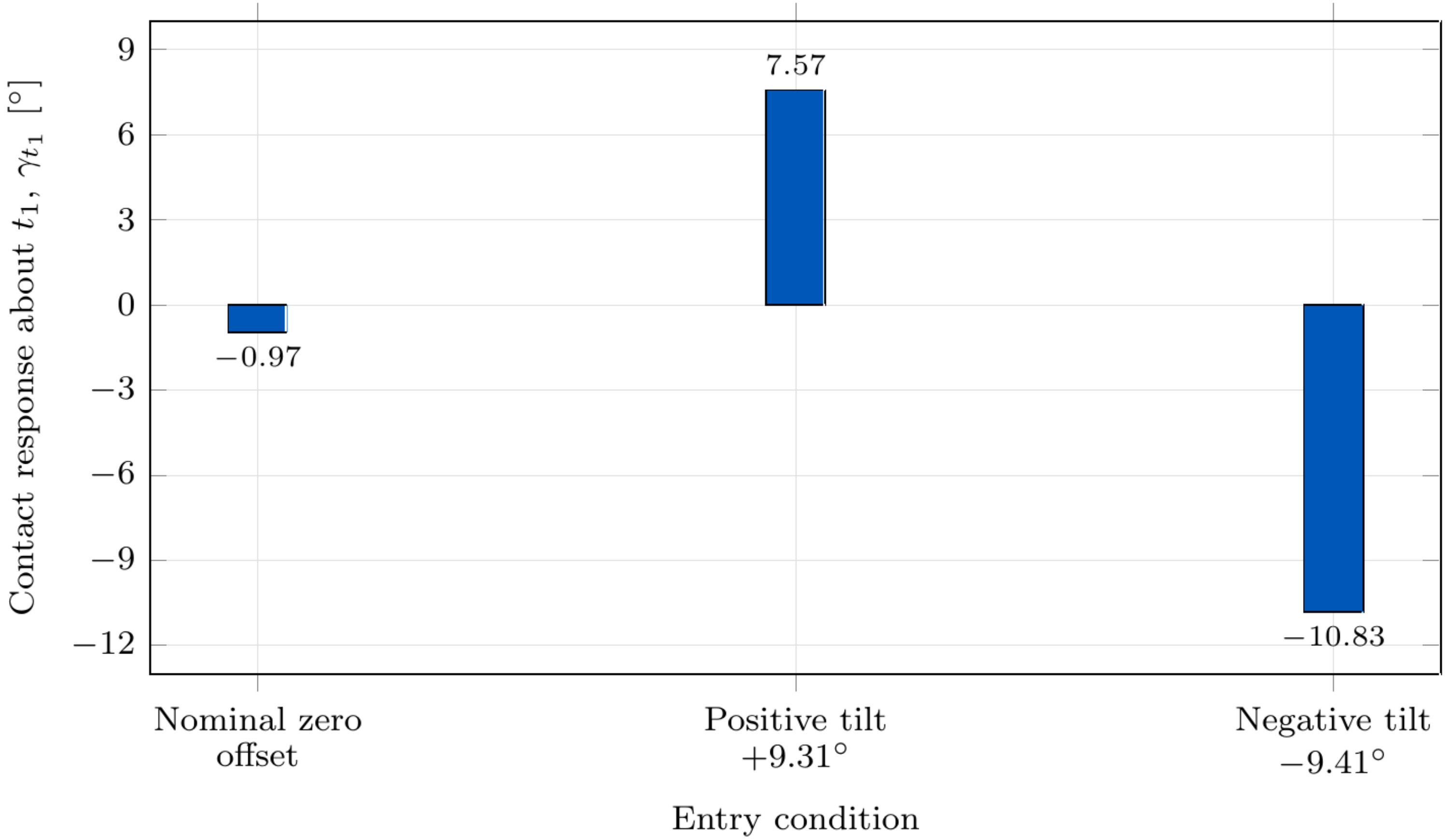
$\theta_{\mathrm{ach},t_1}$ Achieved tilt
Orientation at contact entry

γ_{t_1} Contact response
Rotation relative to the held entry orientation

Aligning response: same sign as the entry tilt

Calculated from measured end-effector orientation

Baseline contact response

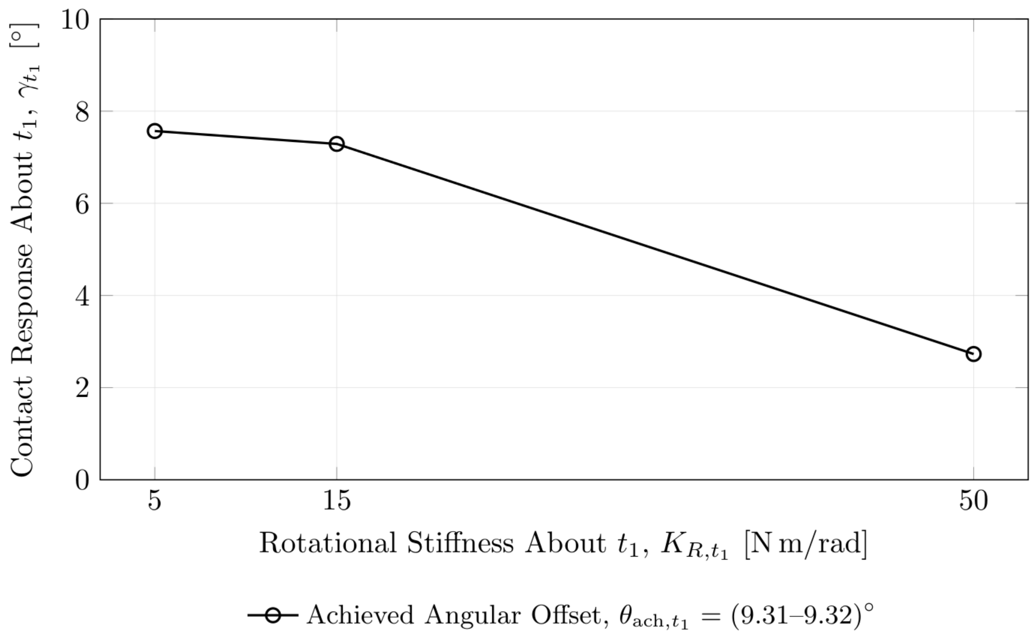


- Centre of compliance at the TCP
- Alignment response for both tilt signs
- Unequal response magnitudes

Effect of stiffness

Rotational stiffness

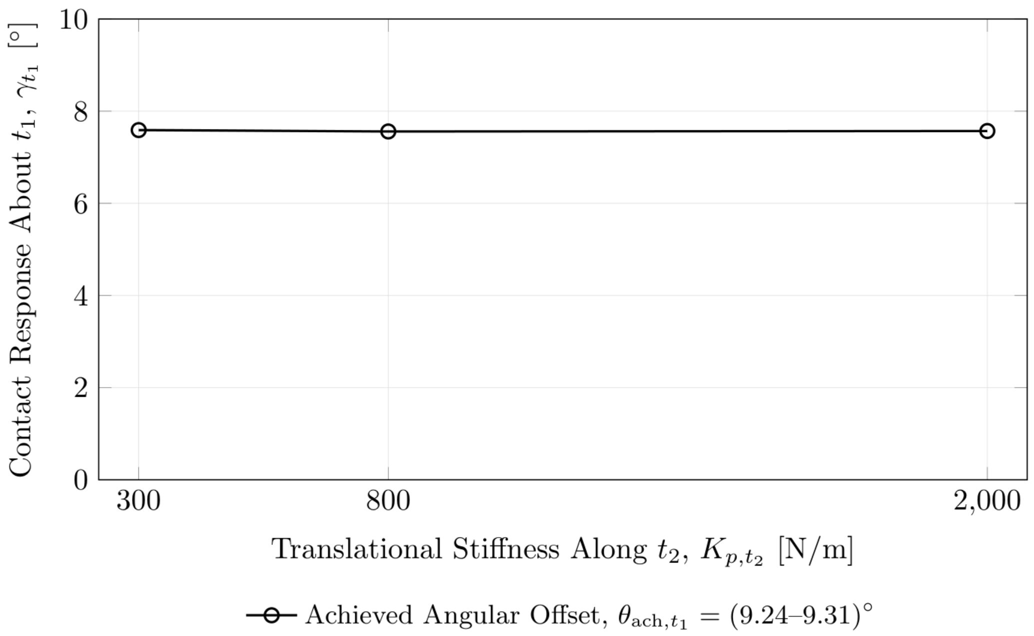
Resistance to rotation about t_1



- **Rotational stiffness: 63.9% lower response**

Translational stiffness

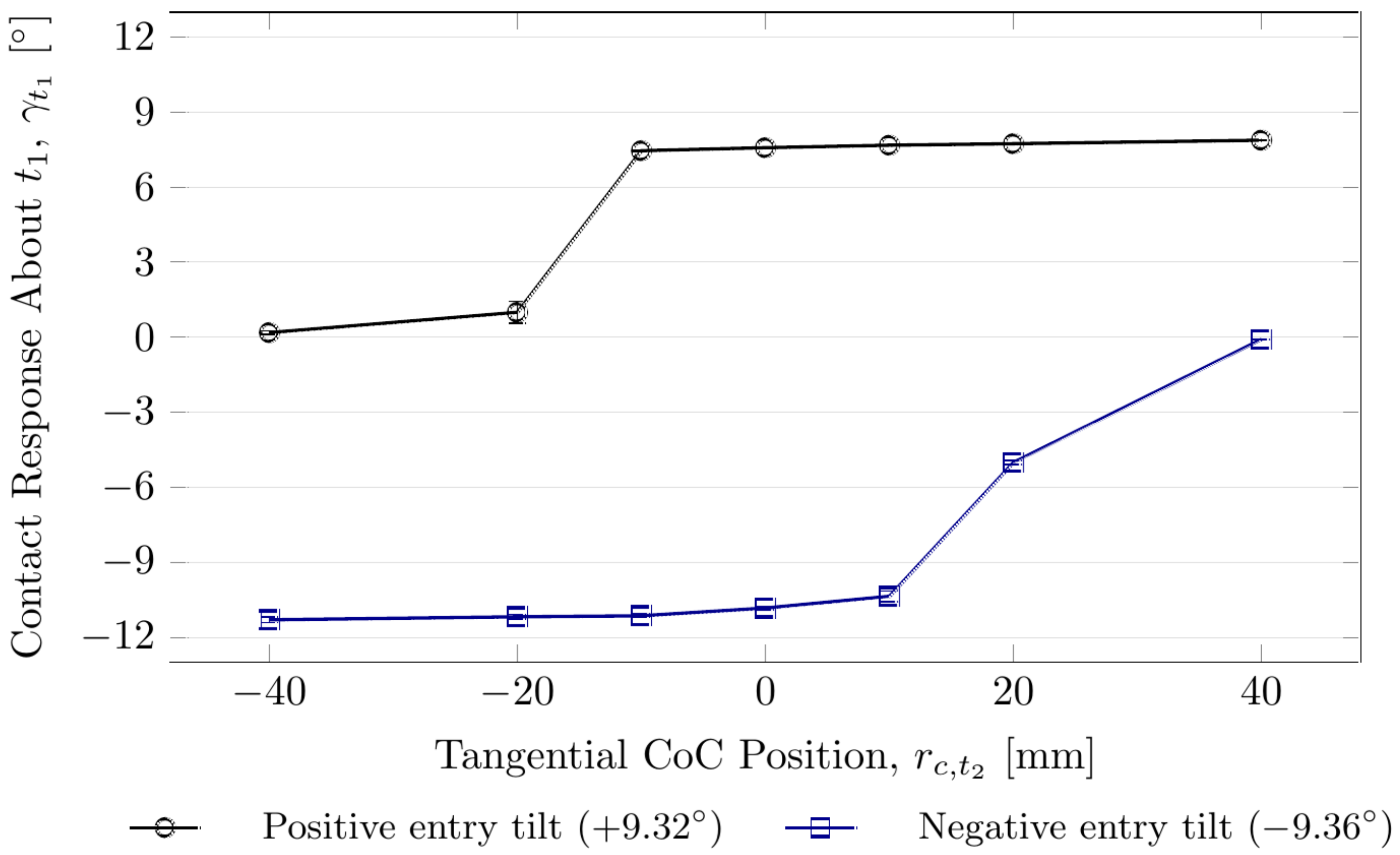
Resistance to sliding along t_2



- **Tangential stiffness: 0.4% response span**

Effect of CoC position

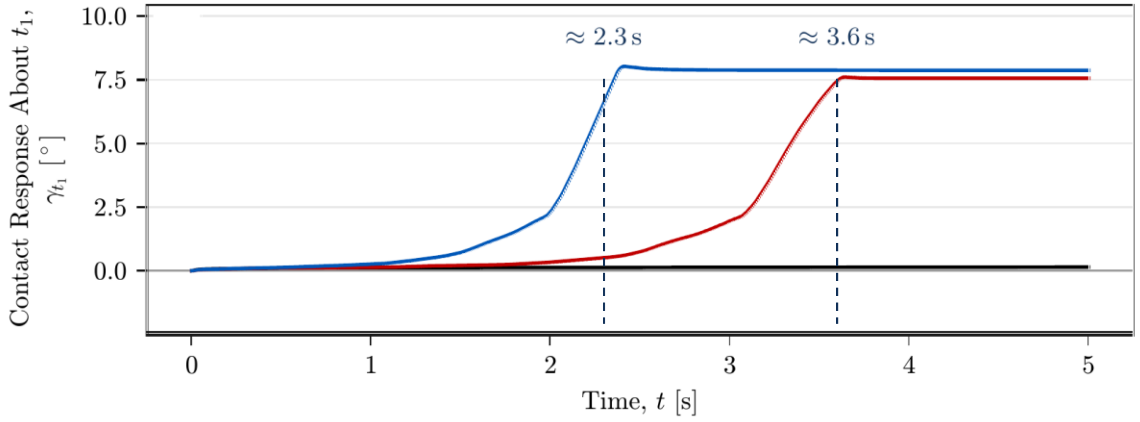
r_{c,t_2} CoC displacement along t_2



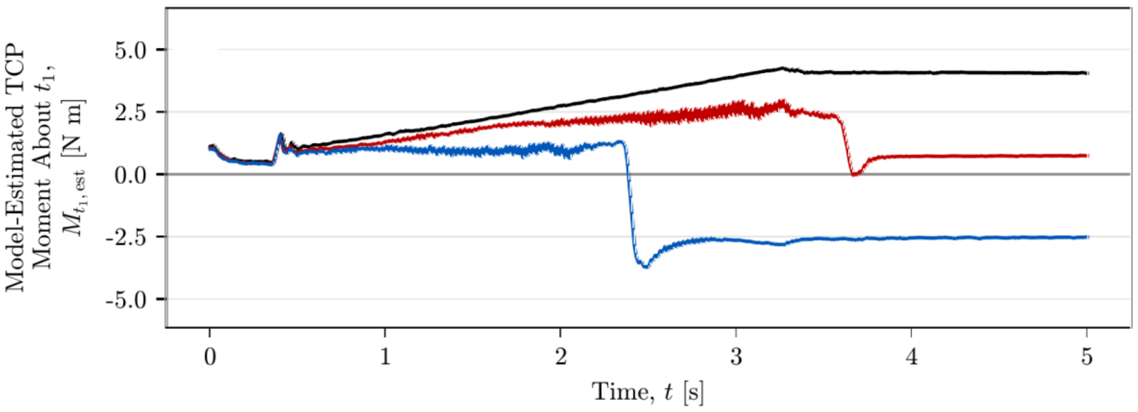
Relative to the TCP-centred baseline

- Supporting displacement: about 4% higher response
- Opposite side: 98–99% lower response
- Required side reverses with the entry tilt

Contact response



Interaction moment at the TCP

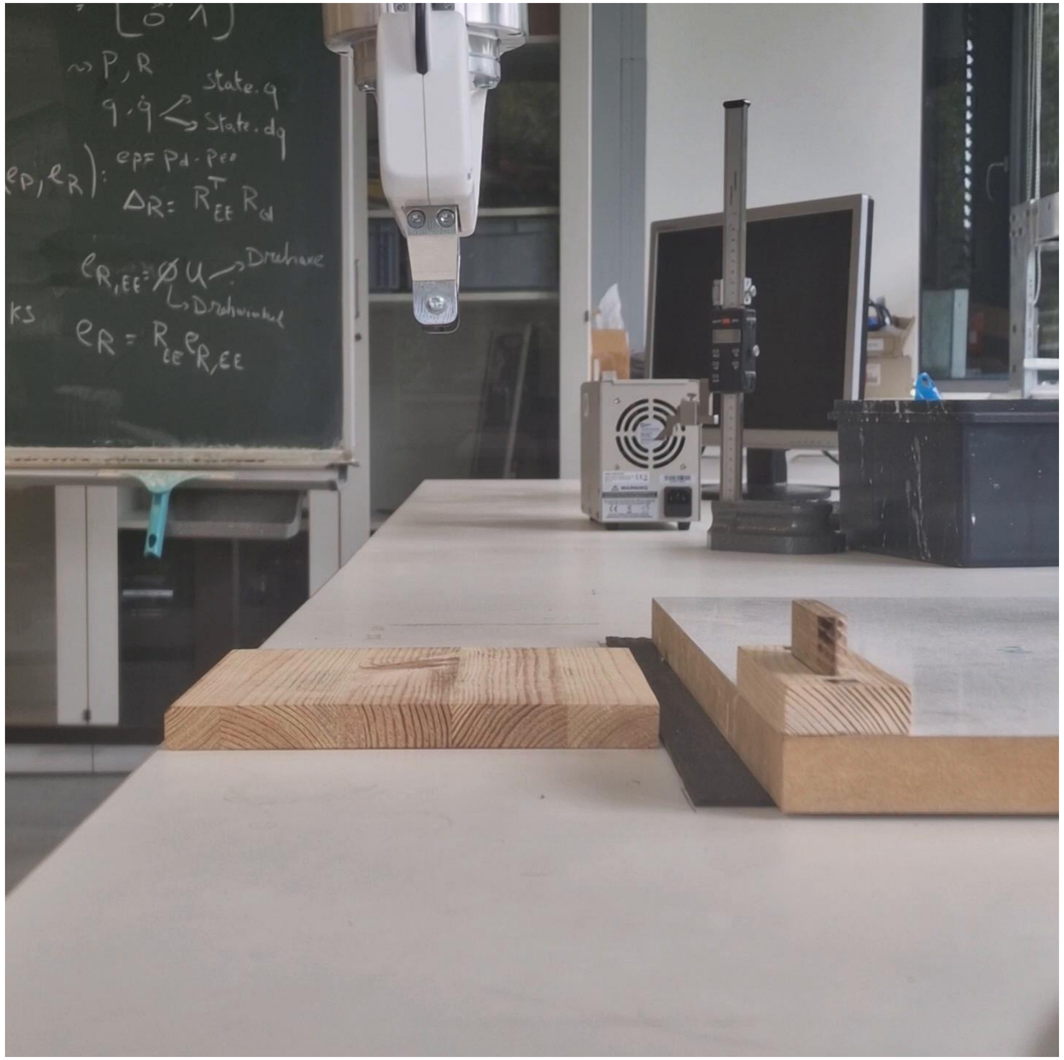


CoC Position, $r_{c,t_2} = -40\,\mathrm{mm}$ CoC at TCP, $r_{c,t_2} = 0$ CoC Position, $r_{c,t_2} = 40\,\mathrm{mm}$

- Reversing CoC displacement reverses the interaction moment
- Supporting displacement: about 1.3 s earlier in these representative traces

Final normal force: −83 N (−40 mm), −79 N (TCP), −78 N (+40 mm), approximately

Contact demonstration

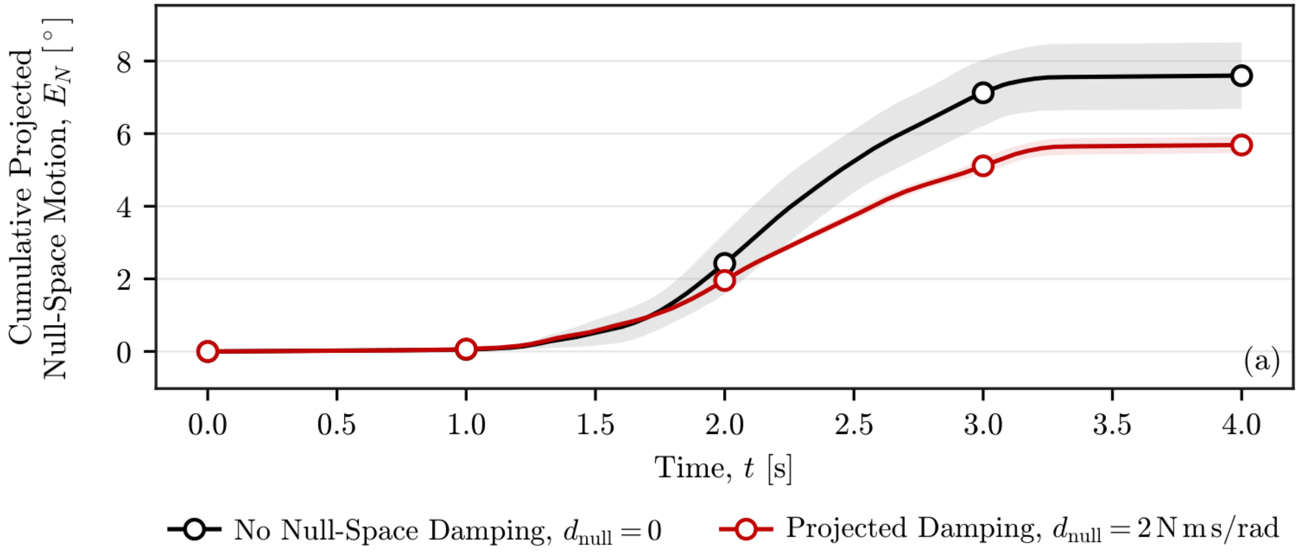


Pre-grinding hold · fixed desired orientation · CoC at TCP

7 joints, 6-DOF tool pose: coordinated motion with the pose held

Damping

Accumulated projected motion



25% less accumulated motion

Maximum TCP position error: 1.304 mm (criterion: 2 mm)

Conditioning

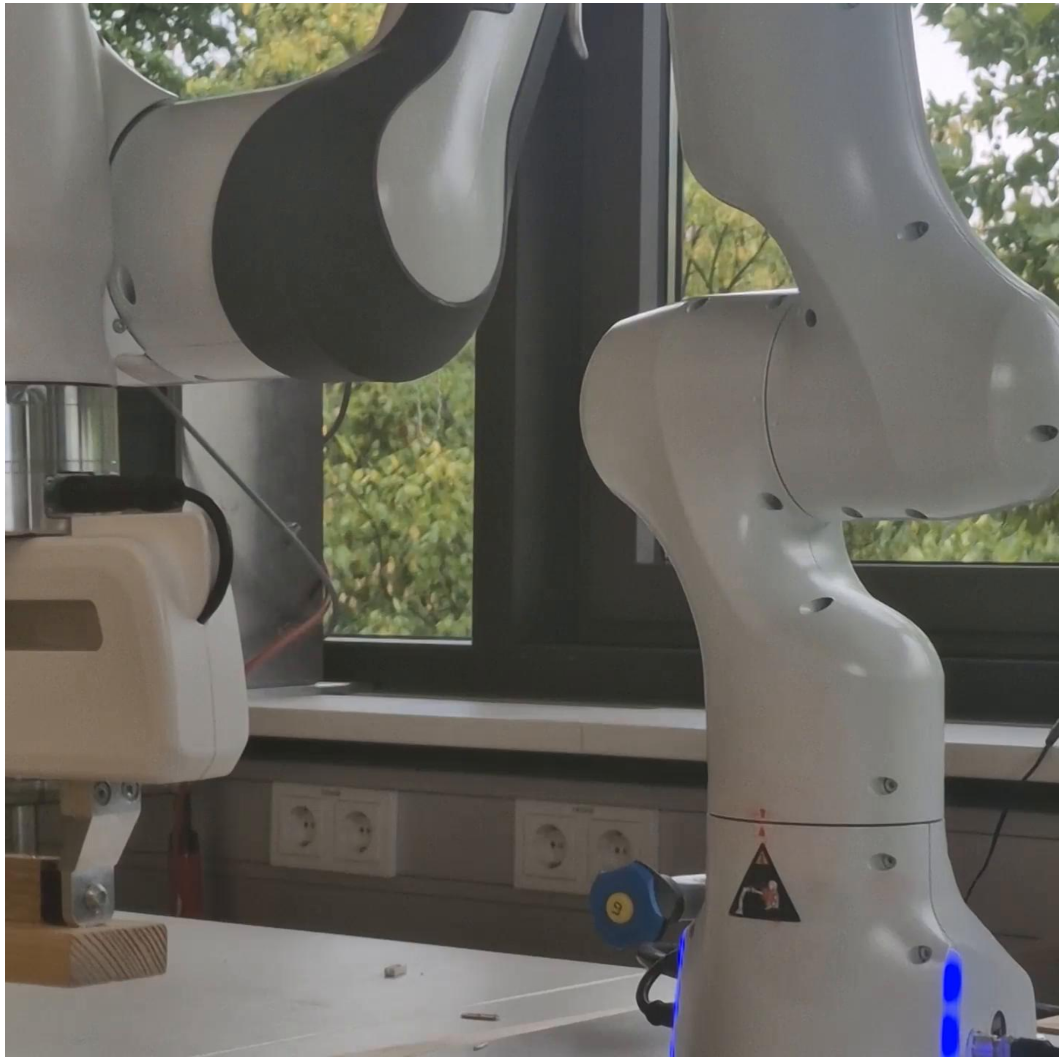
Mean net projected displacement

Setting	Mean (°)
No null-space torque	7.517
Damping	5.609
Conditioning, gain 1.5	0.015
Conditioning, gain 2	−0.006

Near-zero net displacement

Stronger gain: more back-and-forth motion

Disturbance demonstration



No null-space torque → Damping → Conditioning

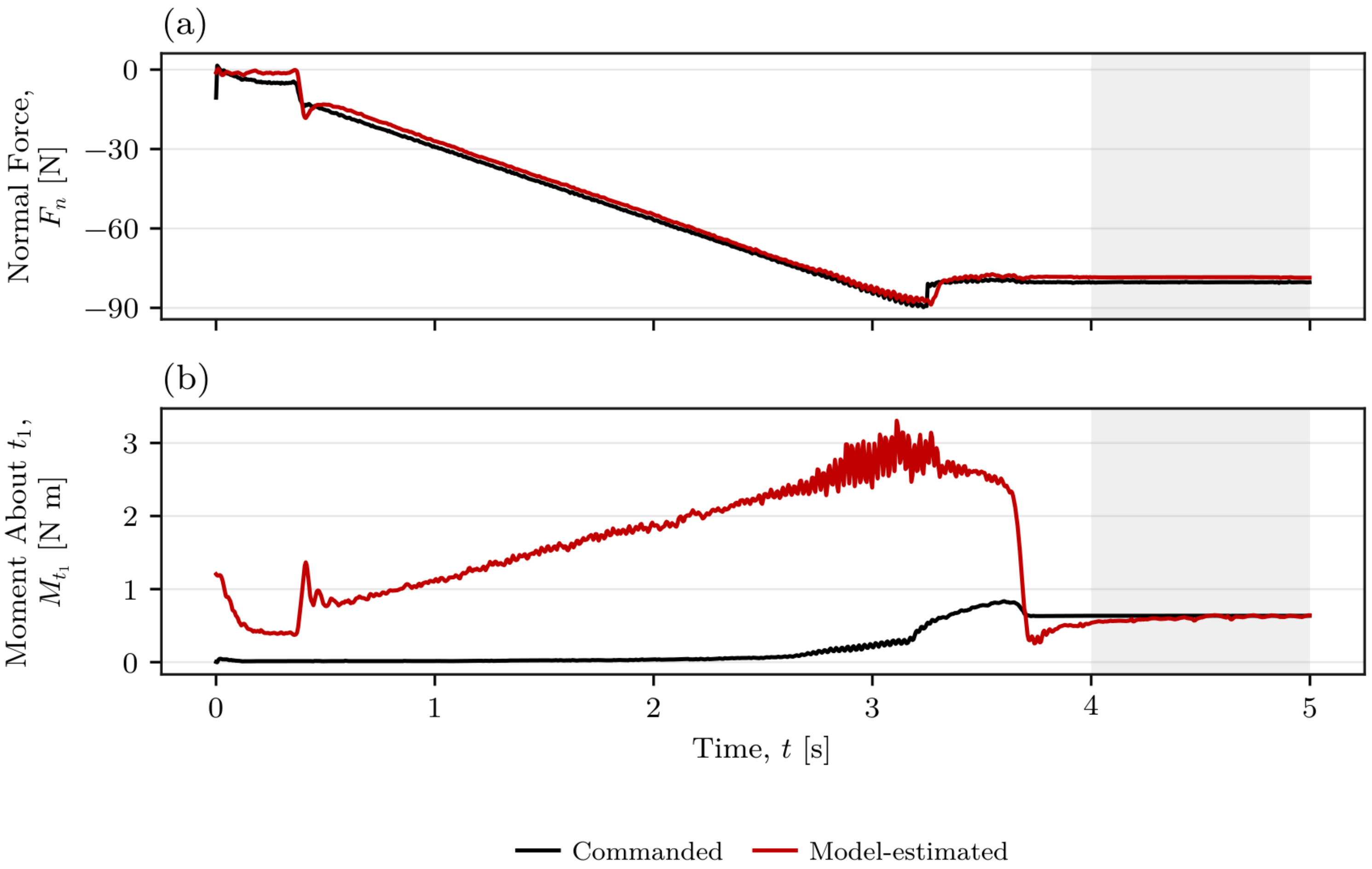
Conclusion

- Passive alignment through Cartesian impedance
- Rotational stiffness controls response magnitude
- CoC displacement selects a preferred rotation direction

Future work

- Tilt-dependent CoC selection
- Return to the TCP after alignment
- Direct angle measurement and sustained grinding

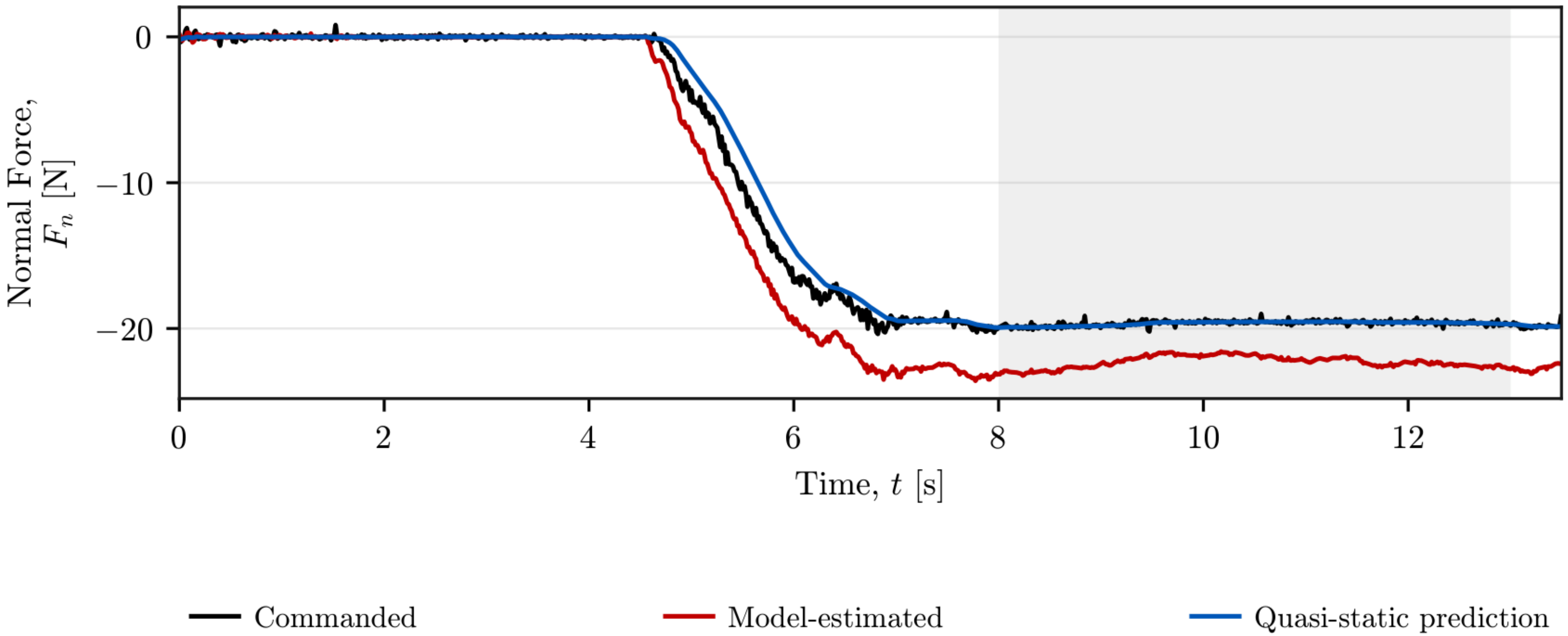
Commanded and estimated wrench



- Stationary force difference: 2.32%
- Stationary moment difference: 4.87%
- Supplementary plausibility evidence

Normal-force plausibility assessment

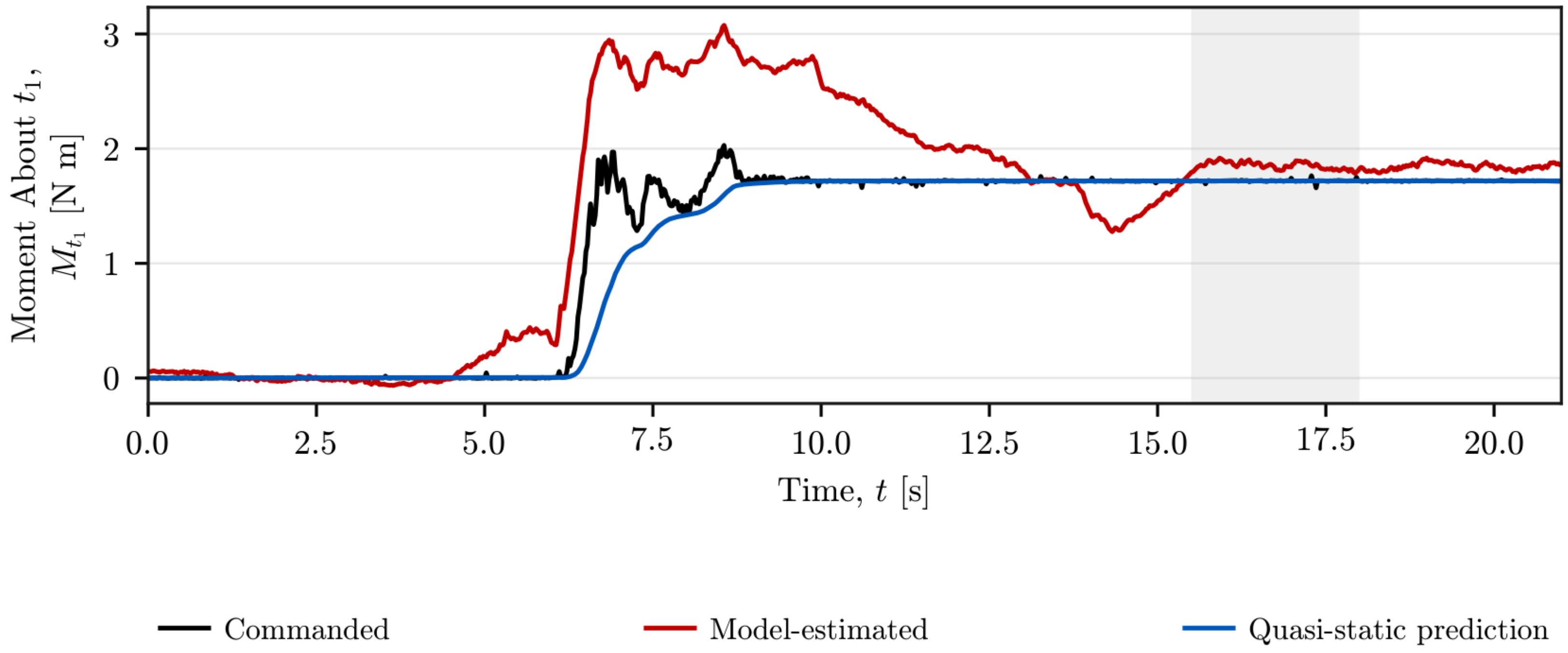
$$F_{n,\mathrm{qs}} = K_{p,n}\, n_s^\top (e_p - e_{p,0})$$



- Spring vs command: 0.05% difference
- Estimate vs command: 13.50% difference

Moment plausibility assessment

$$M_{t_1,\mathrm{qs}} = K_{R,t_1}\, t_1^\top (e_R - e_{R,0})$$



- Spring and commanded mean: 1.719 N m
- Model-estimated mean: 1.852 N m

The EE frame is attached to the controlled tool centre point (TCP).

Position

$$p_{\mathrm{EE}} \in \mathbb{R}^3$$

3 translational degrees of freedom
TCP position in the robot base frame

Orientation

$$R_{\mathrm{EE}} \in SO(3)$$

3 rotational degrees of freedom
EE rotation relative to the robot base

Measured pose

$$T_{\mathrm{EE}} = \begin{bmatrix} R_{\mathrm{EE}} & p_{\mathrm{EE}} \\ 0^\top & 1 \end{bmatrix}$$

Obtained directly from the robot state for position and orientation feedback.

Null-space kinematics

7 joint coordinates control a 6-DOF tool pose: 3 position + 3 orientation.

$$\dim(\ker J) = 7 - \operatorname{rank}(J) = 7 - 6 = 1$$

At a nonsingular configuration: one independent redundant motion direction.

This freedom is a coordinated motion of the joints, not one designated joint.

$$J(q)\,\dot{q}_{\text{null}} = 0, \qquad J \in \mathbb{R}^{6\times 7}$$

J maps joint velocity to tool velocity: null-space motion gives zero tool velocity.

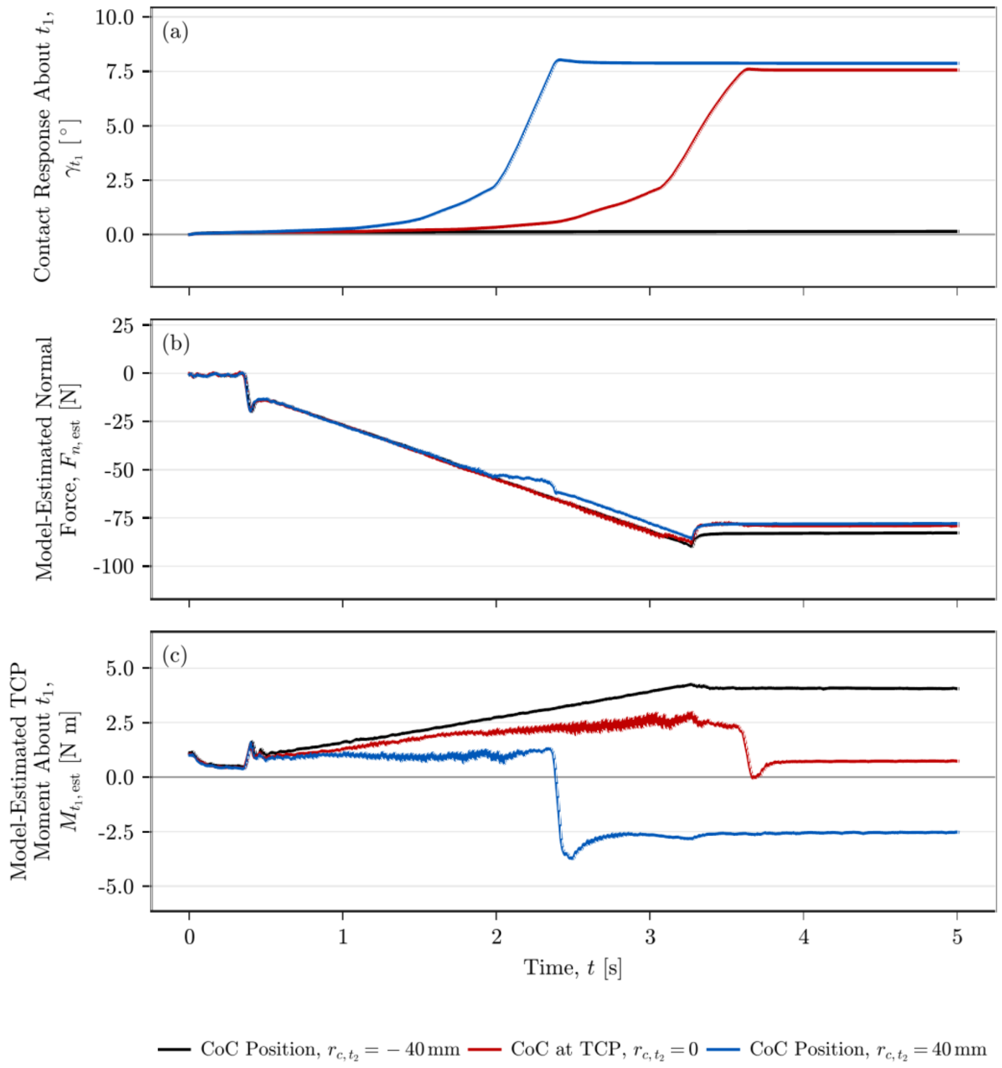
Two torque components control this internal motion:

$$\tau_{\text{null}} = \tau_d + \tau_\sigma$$

Damping torque: counters null-space joint motion.

Conditioning torque: steers towards a configuration with better motion capability.

Contact response, normal force and moment



Representative traces
Positive entry tilt

Similar final normal force

Moment reverses with CoC displacement

Force and moment: model-based robot estimates